INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

15 September 2023 (am)

Subject CS1 – Actuarial Statistics
Core Principles

Paper A

Time allowed: Three hours and twenty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

If you encounter any issues during the examination please contact the Assessment Team on T. 0044 (0) 1865 268 873.

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1 Consider a random variable, X, with a discrete uniform distribution on the integer numbers 20, 21, …, 79.

(i) Determine the expected value of X. [2]

(ii) Determine the standard deviation of X. [3]

[Total 5]

2 An analyst has modelled the loss (in units of £1,000) due to a fire in a supermarket using a random variable, X, with density function:

$$f(x) = \begin{cases} a(x-20), & 0 < x < 20 \\ 0, & \text{otherwise} \end{cases}$$

where a is a real number.

(i) Determine the value of a. [2]

(ii) Calculate the conditional probability that the fire loss exceeds £16,000 given that it exceeds £8,000. [3]

[Total 5]

3 The probability mass function of a random variable, Y, is defined as:

$$P(Y = y) = \binom{k + y - 1}{y}(1 - p)^y p^k, \qquad y = 0, 1, 2, 3, \ldots$$

where k is a positive integer.

(i) State the distribution of the random variable Y. Your answer should include an explanation of the meaning of the distribution's parameters. [2]

(ii) Identify which **one** of the following options gives the natural parameter θ, the scale parameter φ, and the relevant functions $b(\theta)$, $a(\varphi)$ and $c(y, \varphi)$ of the exponential family for this distribution.

A $\theta = \log(1 - p)$, $\varphi = 1$, $b(\theta) = -k \log p$, $a(\varphi) = 1$, $c(y, \varphi) = \log\binom{k + y - 1}{y}$

B $\theta = \log(p)$, $\varphi = 1$, $b(\theta) = -k \log(1 - p)$, $a(\varphi) = 1$, $c(y, \varphi) = \log\binom{k + y - 1}{y}$

C $\theta = \log(1 - p)$, $\varphi = k$, $b(\theta) = -\log p$, $a(\varphi) = k$, $c(y, \varphi) = \log\binom{k + y - 1}{y}$

D $\theta = k \log(1 - p)$, $\varphi = 1$, $b(\theta) = k \log p$, $a(\varphi) = 1$, $c(y, \varphi) = \log\binom{k + y - 1}{k}$.

[3]

(iii) Derive, using the properties of the exponential family, the mean of Y. [2]

(iv) Identify which **one** of the following options gives the variance of Y:

A $V[Y] = -\frac{1}{k}\frac{1-p}{p^2}$

B $V[Y] = \frac{1}{k}\frac{1-p}{p^2}$

C $V[Y] = k\frac{1-p}{p^2}$

D $V[Y] = k\frac{p}{(1-p)^2}$.

[2]

[Total 9]

4 An insurance company collects and maintains a database of key policyholder data in relation to the business that they sell.

(i) List the properties that can lead to this data being classified as 'big data'. [2]

(ii) List two key issues related to the use of data that should be considered by this insurance company. [1]

[Total 3]

5 X and Y are independent random variables. Let W and Z be the random variables defined as follows:

$$Z = \min(X, Y), \qquad W = \max(X, Y)$$

that is, Z gives the smaller and W the larger of the observations of X and Y.

Let F_W, F_X, F_Y and F_Z denote the cumulative distribution functions of W, X, Y and Z, respectively.

(i) Identify which **one** of the following expressions is correct for $F_W(s)$:

A $F_W(s) = F_X(s)\,F_Y(s)$

B $F_W(s) = F_X(s) - F_Y(s)$

C $F_W(s) = F_Y(s) - F_X(s)$

D $F_W(s) = \frac{F_X(s)}{F_Y(s)}$.

[2]

(ii) Show that $F_Z(s) = F_X(s) + F_Y(s) - F_X(s)F_Y(s)$. [5]

The random variable X has an exponential distribution with parameter 6, and Y, independently, also has an exponential distribution with parameter 6.

(iii) (a) Identify which **one** of the following expressions gives the cumulative distribution function of Z:

A $1 - e^{12s}$

B $1 - e^{-12s}$

C $1 - e^{-6s}$

D $1 - e^{-3s}$.

[3]

(b) State the distribution and mean of Z, using your answer to part (iii)(a).

[1]

[Total 11]

6 A railway company investigates the impact of train delays on connections that passengers experience. In a survey, passengers are asked how many trains they used for their journey and whether they have experienced any delays.

Number, x, of trains involved in a journey	1	2	3 or more
Proportion of journeys with x trains that are delayed (in %)	5	12	17
Proportion of all journeys (in %)	60	30	10

In other words, the table states that 60% of all journeys involved only one train. Of those journeys, 5% were delayed. 30% of all journeys involved two trains, and 12% of those journeys were delayed. And finally, 10% of all journeys involved three or more trains, and of those journeys, 17% were delayed.

Using the above survey data:

(i) Calculate the probability that a randomly chosen journey involved fewer than three trains. [1]

(ii) Verify that the probability that a randomly chosen journey was delayed is 8.3%. [2]

(iii) Calculate the probability that a randomly chosen delayed journey involved fewer than three trains. [3]

(iv) Comment on your answers to parts (i) and (iii). [2]

[Total 8]

7 Total losses in a particular company are modelled by a random variable Y with density function:

$$f(y) = \begin{cases} \frac{c}{y^{c+1}}, & y > 1, \quad c > 0 \\ 0, & \text{otherwise.} \end{cases}$$

An analyst wishes to estimate the unknown parameter c.

(i) Identify which **one** of the following expressions gives the maximum likelihood estimate for parameter c:

A $\hat{c} = \frac{c}{\sum_1^n \log(y_i)}$

B $\hat{c} = \frac{cn}{\sum_1^n \log(y_i)}$

C $\hat{c} = \frac{n}{\sum_1^n \log(y_i)}$

D $\hat{c} = \frac{n+1}{\sum_1^n \log(y_i)}$.

[3]

The analyst assumes a gamma prior distribution for c with parameters (a, b).

(ii) Determine the posterior distribution of c with all its parameters. [6]

(iii) Comment on the relationship between the prior distribution and the posterior distribution of c. [1]

(iv) Determine the Bayesian estimate of parameter c under quadratic loss. [2]

[Total 12]

8 The time spent (in minutes) queuing in a line by customers at a bank is modelled by the random variable X with density $f(x) = \theta x e^{\frac{-\theta x^2}{2}},\ x > 0$. The parameter θ is assumed to follow a priori a gamma distribution with parameters a and b.

(i) Identify which **one** of the following expressions is proportional to the posterior density of θ given a random sample $x_1, x_2, \ldots x_n$ of observations from X:

A $\quad f(\theta|x) \propto \theta^{n+a} e^{-\theta\left(\frac{\sum_{i=1}^{n} x_i^2}{2}+b+1\right)}$

B $\quad f(\theta|x) \propto \theta^{n+a-1} e^{-\theta\left(\frac{\sum_{i=1}^{n} x_i^2}{2}+b\right)}$

C $\quad f(\theta|x) \propto \theta^{n+a-2} e^{-\theta\left(\frac{\sum_{i=1}^{n} x_i^2}{2}+b\right)}$

D $\quad f(\theta|x) \propto \theta^{n+a-1} e^{-\theta\left(\frac{\sum_{i=1}^{n} x_i^2}{2}+b-1\right)}$.

[3]

(ii) State the posterior distribution of θ in part (i) and its parameters. [3]

The waiting times of ten customers are recorded in the table below and the prior knowledge is determined with $a = 4, b = 1.5$.

2.5	1.25	3	1.5	5.5	4	3.25	2	1.5	1

(iii) Calculate a point estimate of the time spent queuing in the line using Bayesian estimation under all-or-nothing loss. [4]

[Total 10]

9 The following nine pairs of data are given on observations from two random variables X and Y:

x	5	7	0	6	8	1	4	9	2
y	11.00	16.18	0.68	11.99	17.72	2.91	9.64	18.92	6.31

Summary statistics of the data are shown below:

$$\sum x_i = 42, \sum y_i = 95.35, \sum x_i^2 = 276, \sum y_i^2 = 1340.194, \sum x_i y_i = 606.33$$

(i) Perform a suitable statistical test to investigate the hypothesis that Pearson's population correlation coefficient for X and Y is positive, at significance level 0.01. [6]

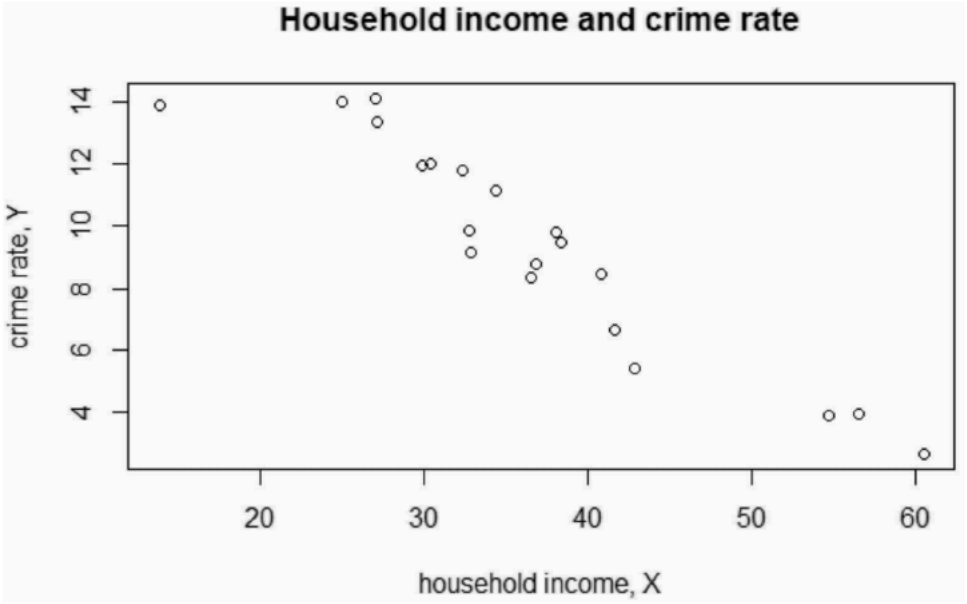
(ii) Comment on the relationship between X and Y, based on your answer in part (i). [2]

(iii) Fit a simple linear regression model of y on x. [3]

(iv) Determine a 99% confidence interval for the predicted mean response, when $x = 3$. [5]

[Total 16]

10 For a socio-economic analysis, a random sample of 20 regions is considered. An analyst collects data on average household income (in units of \$1,000, denoted by X) and crime rate (in percent, denoted by Y) for each region in the sample. The data are displayed in the following plot, and summary statistics are given below.



$$\sum_{i=1}^{20} x_i = 733, \sum_{i=1}^{20} y_i = 189, \sum_{i=1}^{20} x_i^2 = 29{,}203, \sum_{i=1}^{20} y_i^2 = 2{,}009, \sum_{i=1}^{20} x_i\, y_i = 6{,}208$$

(i) Calculate a 95% confidence interval for the mean household income. [5]

(ii) Calculate Pearson's correlation coefficient for the relationship between household income and crime rate. [4]

(iii) Justify why Pearson's correlation coefficient is appropriate in this context as compared to alternative correlation coefficient measures. [2]

For the three regions with the highest household income the following data have been observed:

Household income (in units of \$1,000), X	54.73	56.61	60.54
Crime rate (%), Y	3.896	3.958	2.658

(iv) Perform a statistical test to decide if there is a significant difference in the mean crime rate between the group of regions with an average household income of more than \$50,000 and the group of regions with an average household income of less than \$50,000. You can assume that the variances of crime rates are the same in all regions. [8]

(v) Comment on the assumption of equal variances made in part (iv). [2]

[Total 21]

END OF PAPER